4 **(a)** A ball is held between two fixed points A and B by means of two stretched springs as shown in Fig. 4.1.

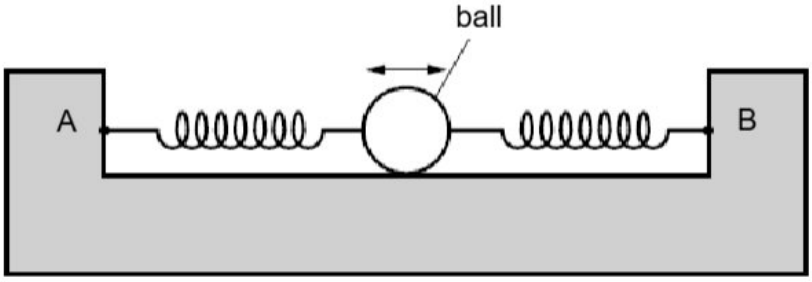


Fig. 4.1

The ball is free to oscillate horizontally along the line AB on the smooth plane.

The variation of the acceleration *a* of the ball with its displacement *x* from its equilibrium position is shown in Fig. 4.2.

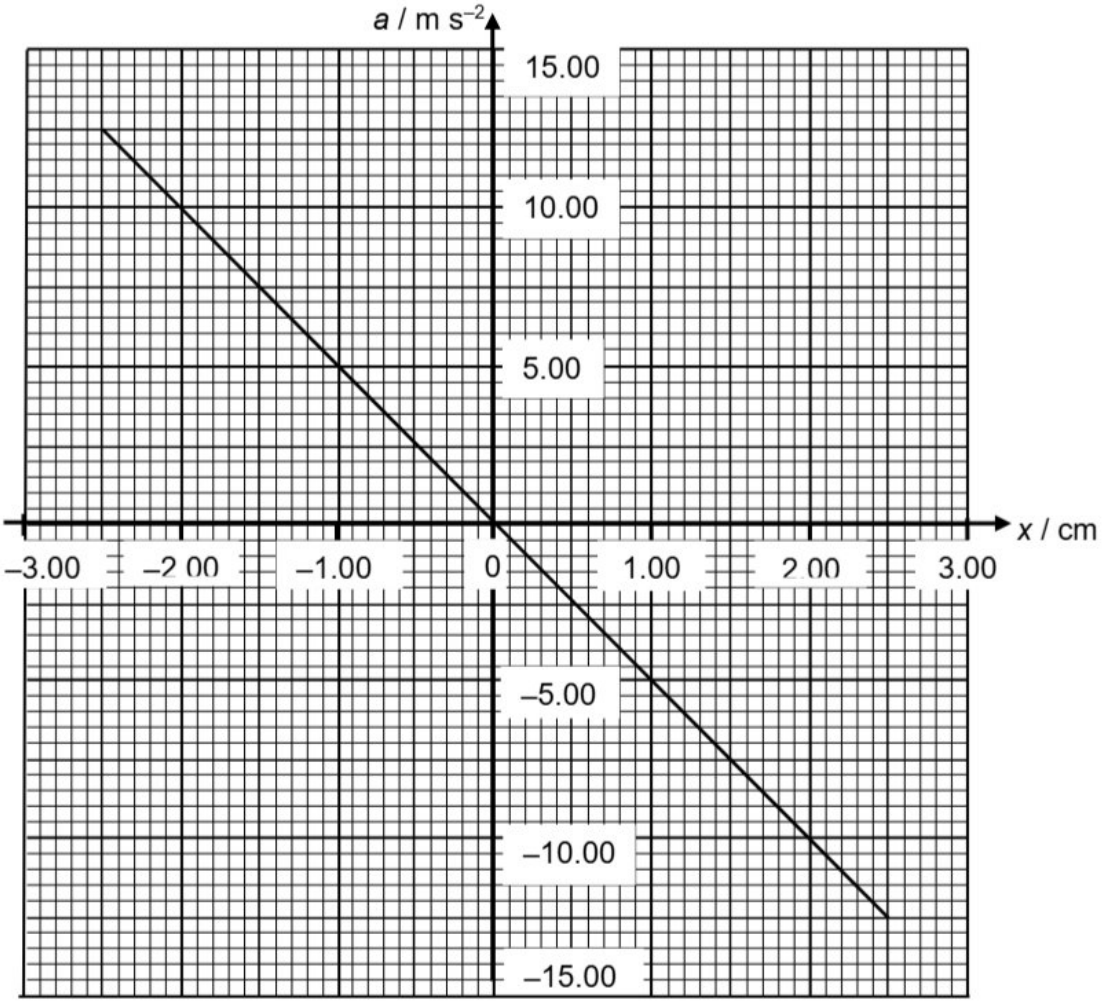


Fig. 4.2

(i) State and explain the features of Fig. 4.2 which indicate that the motion of the ball is simple harmonic.

.....

.....

.....

..... [2]

(ii) On Fig. 4.3, sketch the labelled velocity-displacement graph of the simple harmonic motion which was illustrated in Fig. 4.2. Intersection points on the horizontal and vertical axis need to be indicated in this figure.

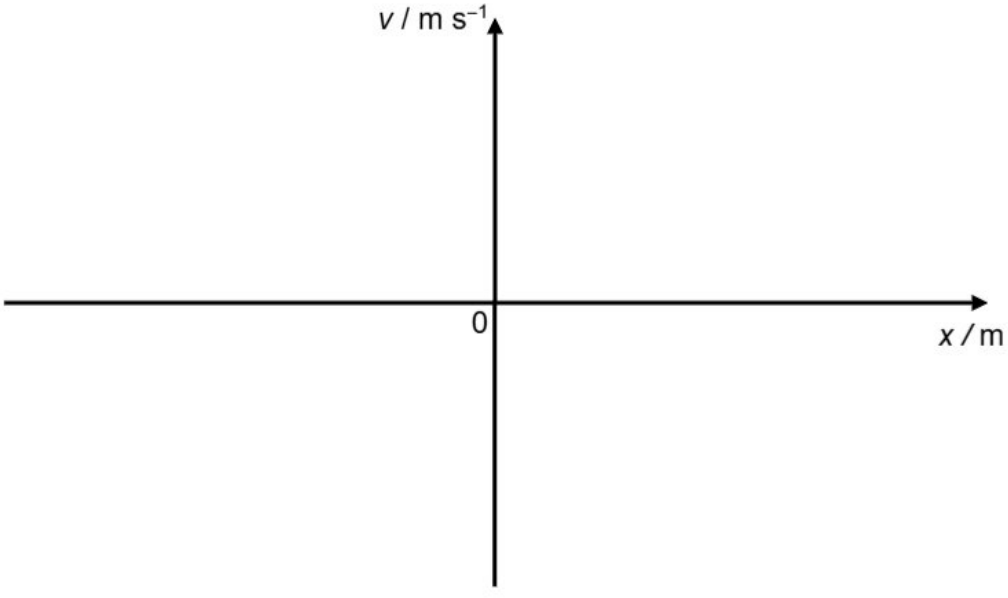


Fig. 4.3

(b) A student removes the ball and attaches it to the apparatus illustrated in Fig. 4.4 in order to investigate the oscillation vertically.

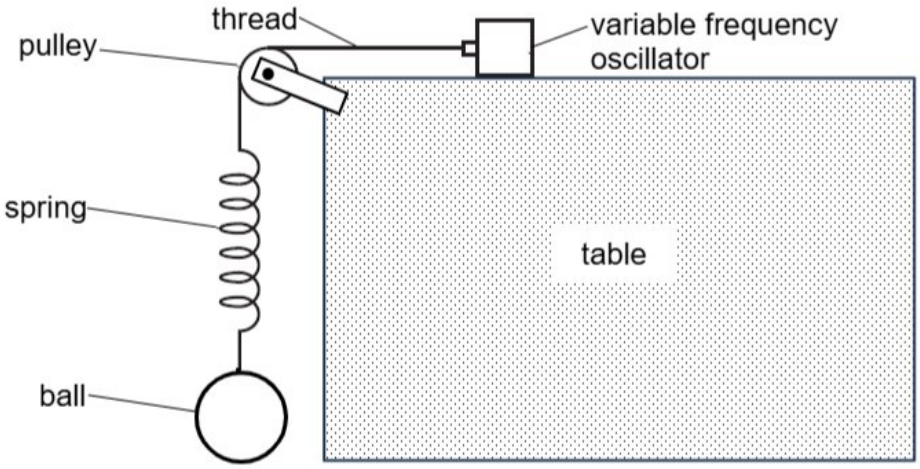


Fig. 4.4

The amplitude of the vibrations produced by the oscillator is constant. The variation with frequency of the amplitude of the oscillations of the ball is shown in Fig. 4.5. The mass of the ball is given to be 150 g, and its oscillations may be assumed to be simple harmonic.

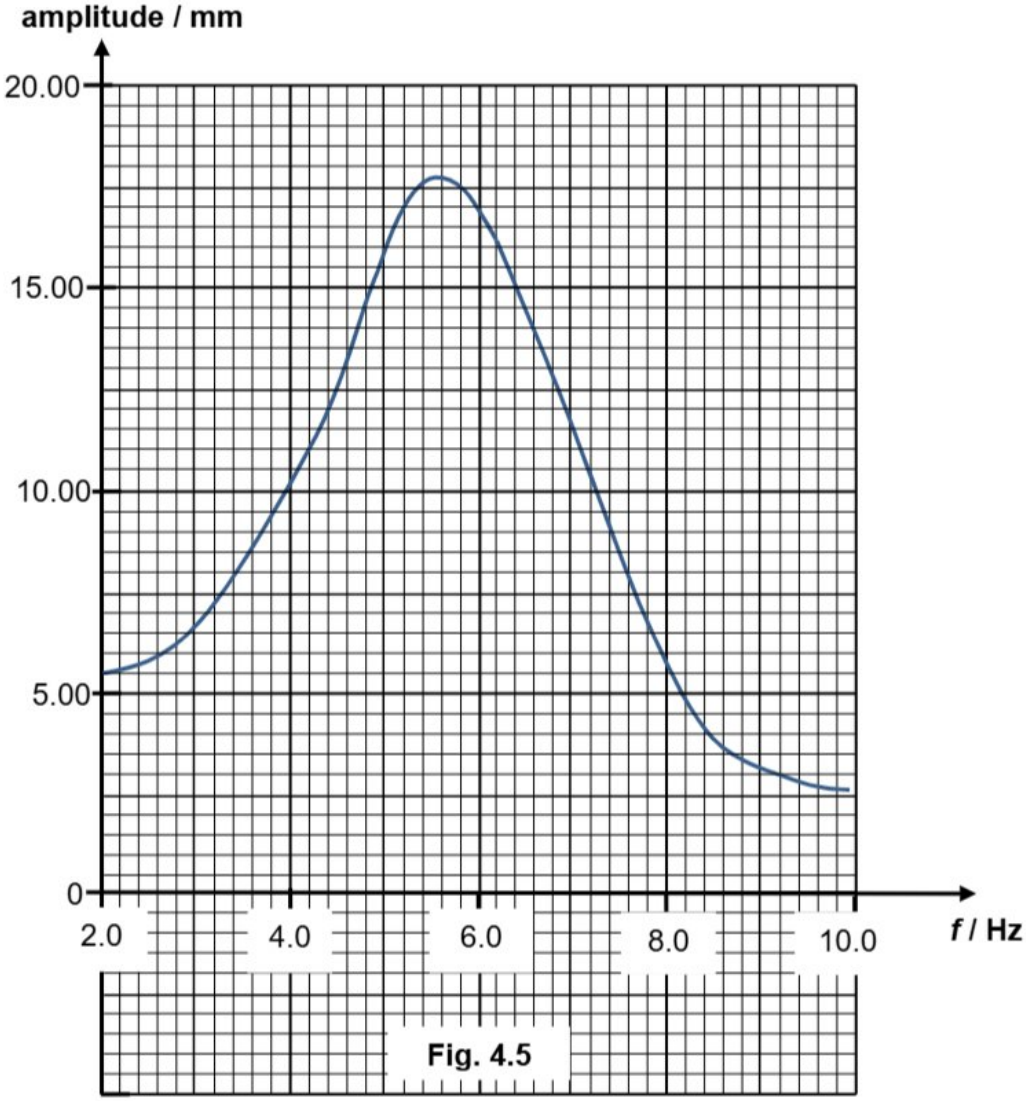


Fig. 4.5

(i) State the amplitude and the frequency when resonance of the system illustrated in Fig. 4.5 occurs.

amplitude = ... mm

frequency = ... Hz [1]

(ii) **1.** Sketch the free body diagram of the ball when the tension of the spring is at its maximum.

[1]

2. Hence, determine the maximum tension in the spring.

tension = ... N [3]

(iii) Some very light but large feathers are attached to the bottom of the ball so that the feathers extend outwards. The investigation is now repeated.

On Fig. 4.5, draw a line to show the new variation with frequency of the amplitude of variation for frequencies between 2 Hz and 10 Hz. [2]